Girls Build Bots — Cadette Coding Basics

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** Coding for Good — Coding Basics **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 2-hour session · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

Cadettes get the real framing: code runs the world they live in — their feeds, their school systems, their music — and the people who write it decide how it behaves. This badge is about stepping from *user* to *builder*.

Vocabulary: algorithm, conditional, variable, loop, event, function (a named chunk of steps), debug, pseudocode.

Coding Basics (2 hours)

Badge theme: Writing programs with real computer-science concepts — conditionals, variables, loops, and events — and understanding how code powers daily technology.

Materials

- Laptops with **MakeCode Arcade** (arcade.makecode.com)
- Projector for live coding; printed pseudocode reference card
- Sticky notes for the code-review swap

Activity 1 — Code Runs Your Day (15 min)

Trace one ordinary hour of a Cadette's day and find the code in it: the alarm (a scheduled event), the playlist shuffle (a randomizing algorithm), the thermostat (a conditional: *if* below 68, heat on). Then the honest kicker: someone *wrote* each of those decisions. Coders decide how technology behaves — that's the power this badge hands over.

Activity 2 — The Four Tools (20 min)

Whiteboard the core toolkit with a physical demo for each: **event** (when the door opens → wave), **conditional** (if I say "blue," stand up), **variable** (a running count on the board), **loop** (repeat until told to stop). Then write one line of **pseudocode** for each as a group — the bridge from idea to program.

Activity 3 — Build: Reaction-Time Tester (55 min)

Progressive build of a genuinely useful program — a reaction-time game: 1. **Event:** on button press, start the round 2. **Variable:** a timer counts up; a score stores the best time 3. **Conditional:** *if* pressed too early → "Too soon!"; *if* new best → update the record 4. **Loop:** play rounds forever, tracking best score 5. Stretch: add a two-player mode with two variables

Every Scout leaves with a working program using all four tools — and it's a real tool, not a toy: reaction time is what sports scientists actually measure.

Activity 4 — Code Review Swap (20 min) + Wrap (10 min)

Swap laptops. Each Scout reads her partner's program and answers on a sticky note: What does it do? Point to a conditional and a variable. One thing you'd improve. Then discuss in pairs and make one improvement to your own. **This is exactly how professional software teams work.** Closing: which of the four tools would you teach a friend first, and how?

> **Robot moment (optional):** Share for 60 seconds where each of the four tools lives in the team's robot code — sensor events, if-statements steering autonomous mode, variables tracking game state, the main control loop.

Completion Checklist — Coding Basics

By the end of the session, to earn the badge, each Scout can: - [ ] Identify code making decisions in everyday technology - [ ] Explain events, conditionals, variables, and loops with examples - [ ] Build a working program that uses all four concepts - [ ] Read a peer's code and give constructive review feedback

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Mentor Guide (for the robotics team)

- **Treat Cadettes as junior colleagues.** Middle schoolers disengage instantly when talked down to; they lean in when handed real tools.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on.
- **Debug out loud.** When you help fix code, narrate your reasoning — hearing the process is the lesson.
- **Let the code review be real.** "I'd improve…" feedback, given and taken seriously, is the most professional 20 minutes of the session.
- **Share a failure story.** "Our robot bricked at competition over one line" is the most credible teaching you can do.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Cadette Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.